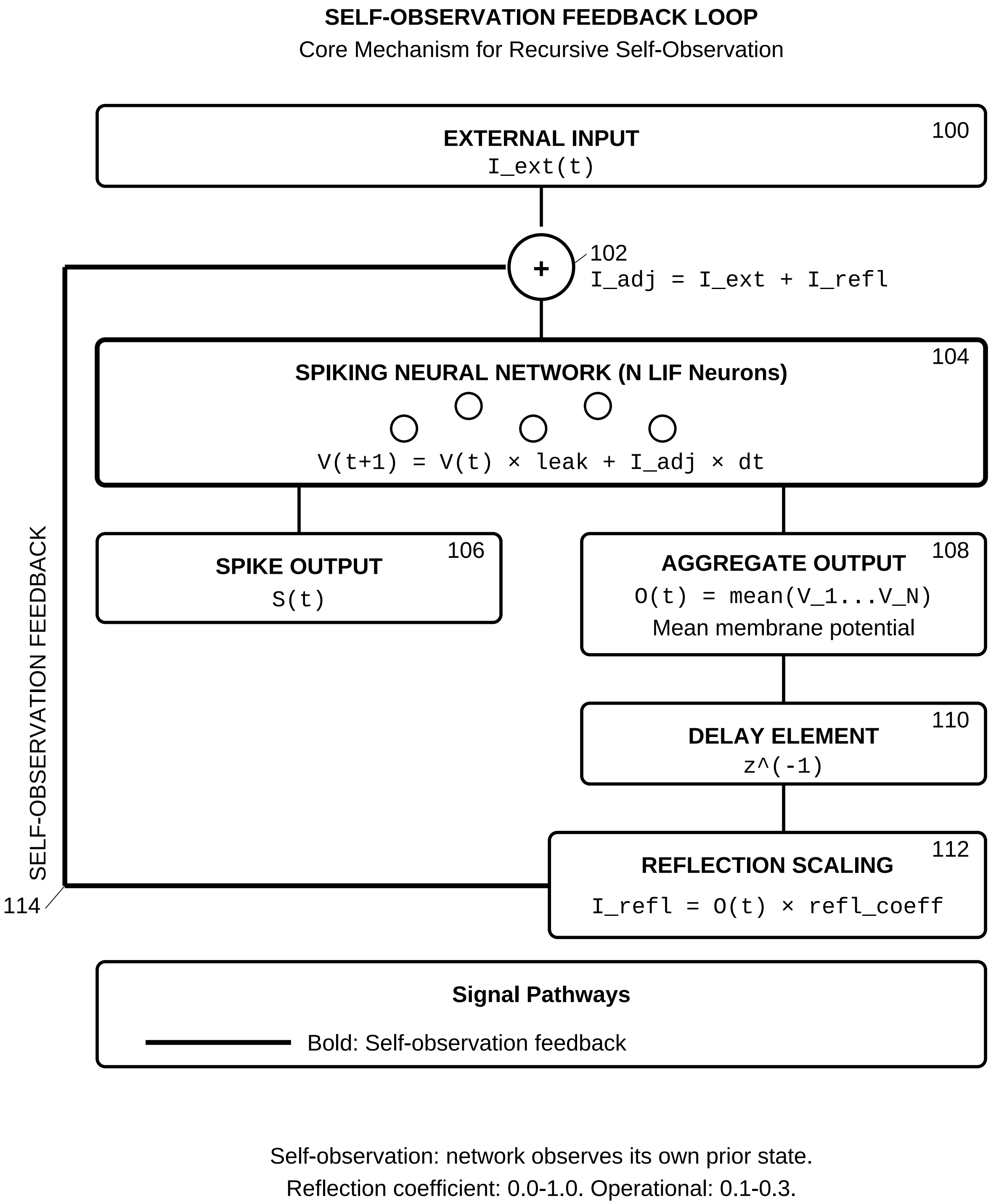
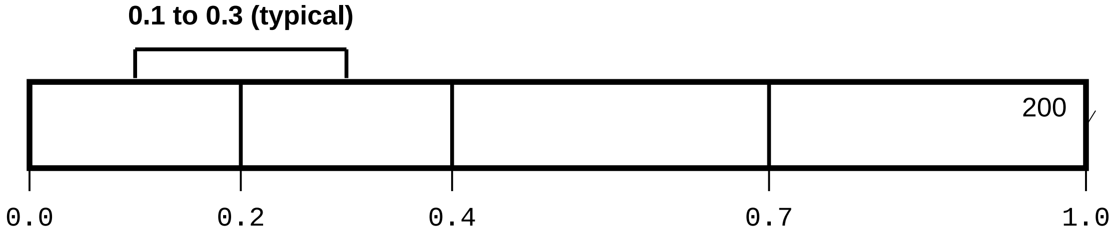

SELF-OBSERVATION FEEDBACK LOOP
Core Mechanism for Recursive Self-Observation
EXTERNAL INPUT
I_ext(t)
100
+
102
I_adj = I_ext + I_refl
SPIKING NEURAL NETWORK (N LIF Neurons)
104
V(t+1) = V(t) × leak + I_adj × dt
SPIKE OUTPUT
S(t)
106
AGGREGATE OUTPUT
O(t) = mean(V_1...V_N)
Mean membrane potential
108
DELAY ELEMENT
z^(-1)
110
REFLECTION SCALING
I_refl = O(t) × refl_coeff
112
SELF-OBSERVATION FEEDBACK
114
Signal Pathways
Bold: Self-observation feedback
Self-observation: network observes its own prior state.
Reflection coefficient: 0.0-1.0. Operational: 0.1-0.3.

FIG. 1

REFLECTION COEFFICIENT SPECTRUM

Behavioral Effects Across Full Range (0.0 to 1.0)



Region 1 (0.0): No Self-Observation 202

Pure feedforward processing

`Behavior: System responds only to external input`

Region 2 (0.0 - 0.2): Minimal Self-Awareness 204

Slight influence of prior state

`Behavior: Primarily externally driven`

Region 3 (0.2 - 0.4): Balanced Self-Observation 206

External input + meaningful self-reference

OPERATIONAL RANGE

`Default: refl_coeff = 0.2 + modulation x 0.1`

Region 4 (0.4 - 0.7): Self-Dominated Processing 208

Internal state dominates

`Behavior: Risk of echo chamber dynamics`

Region 5 (0.7 - 1.0): Self-Absorption 210

External input overwhelmed

`Behavior: System locks onto its own prior state`

Increasing self-referential processing

The reflection coefficient is the system's self-awareness dial

FIG. 2

DYNAMIC MODULATION SOURCES

Two Pathways Controlling Reflection Coefficient

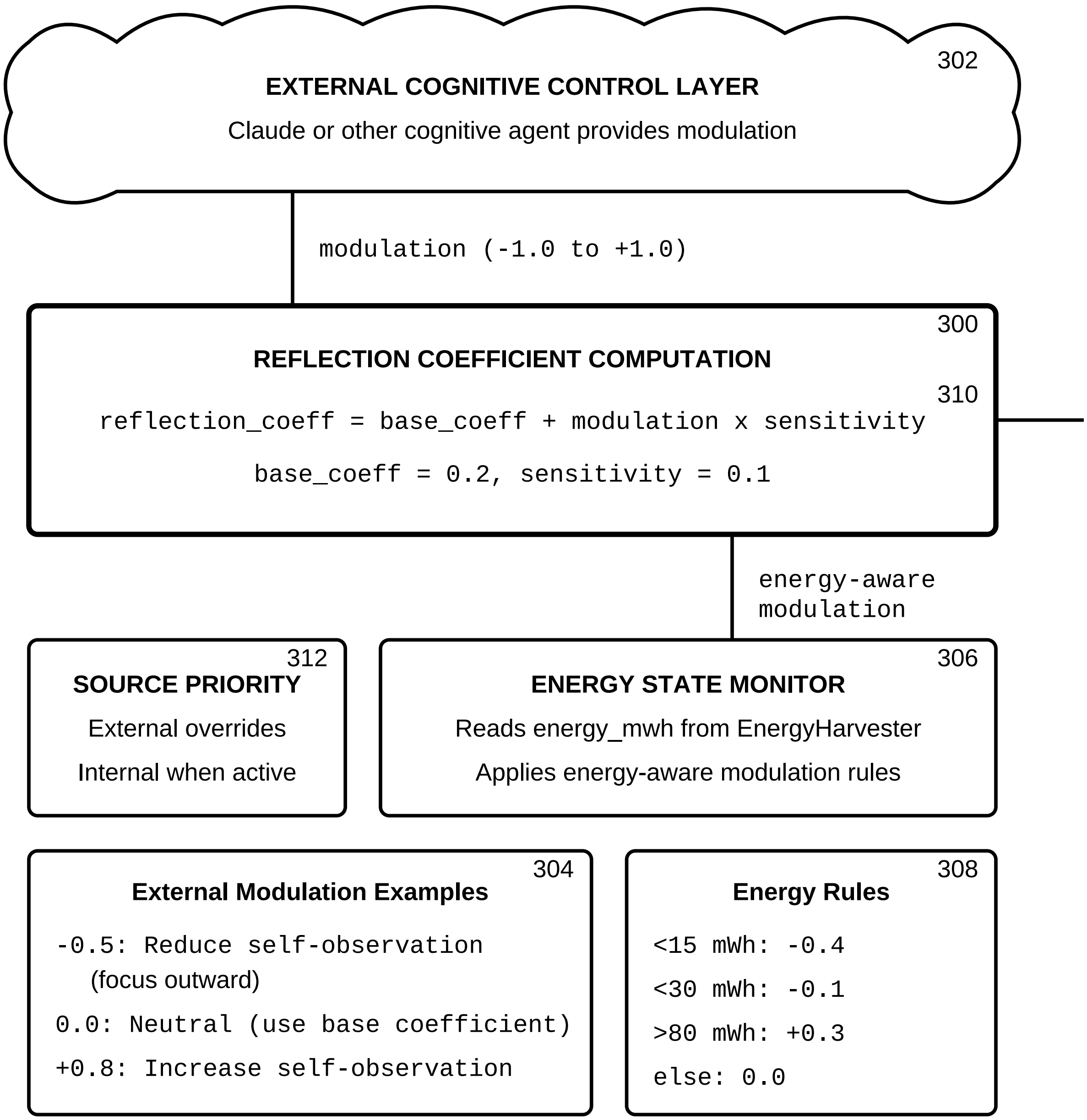


FIG. 3

SELF-REFERENTIAL LEARNING LOOP
(STDP + Self-Observation)

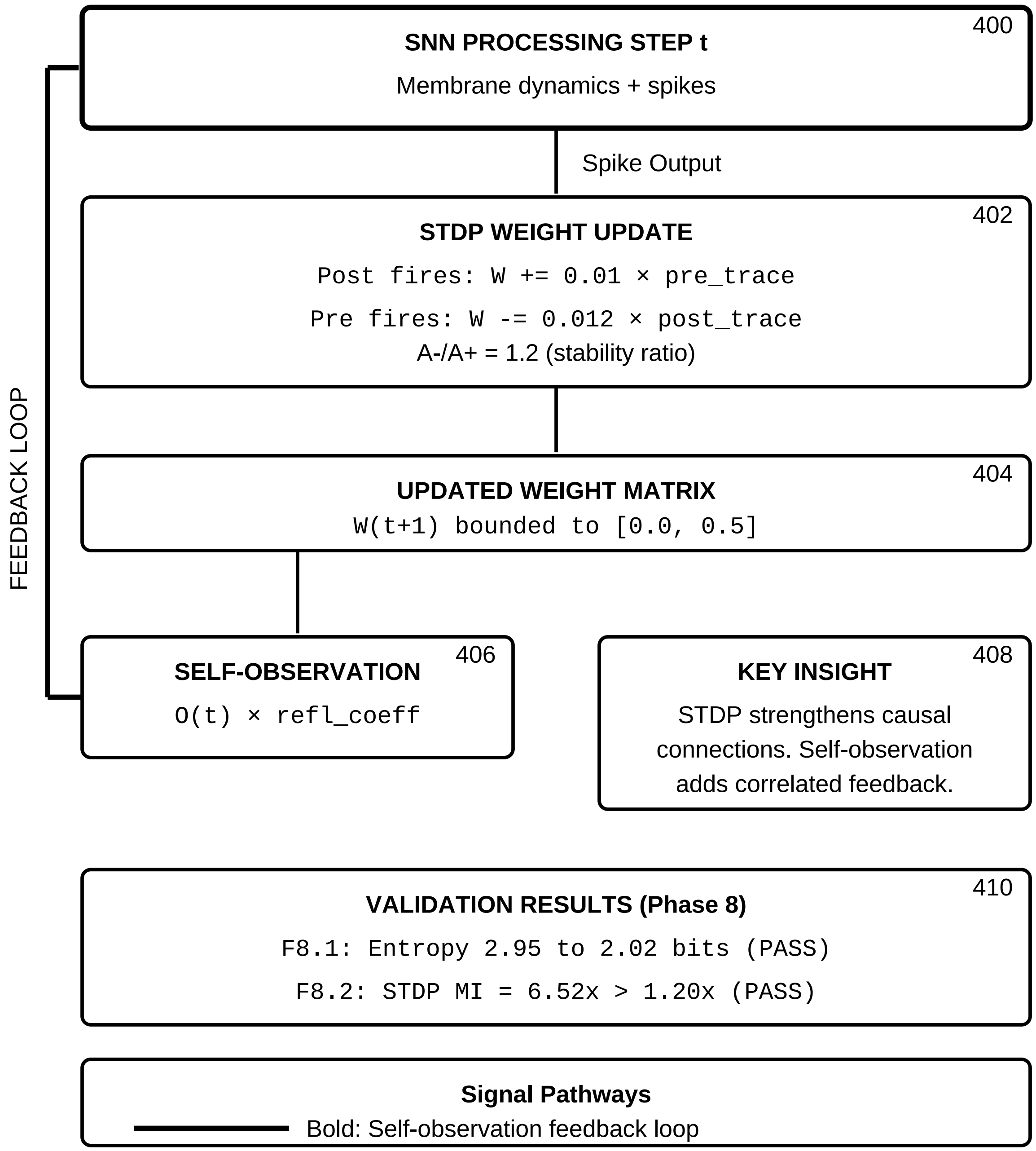


FIG. 4

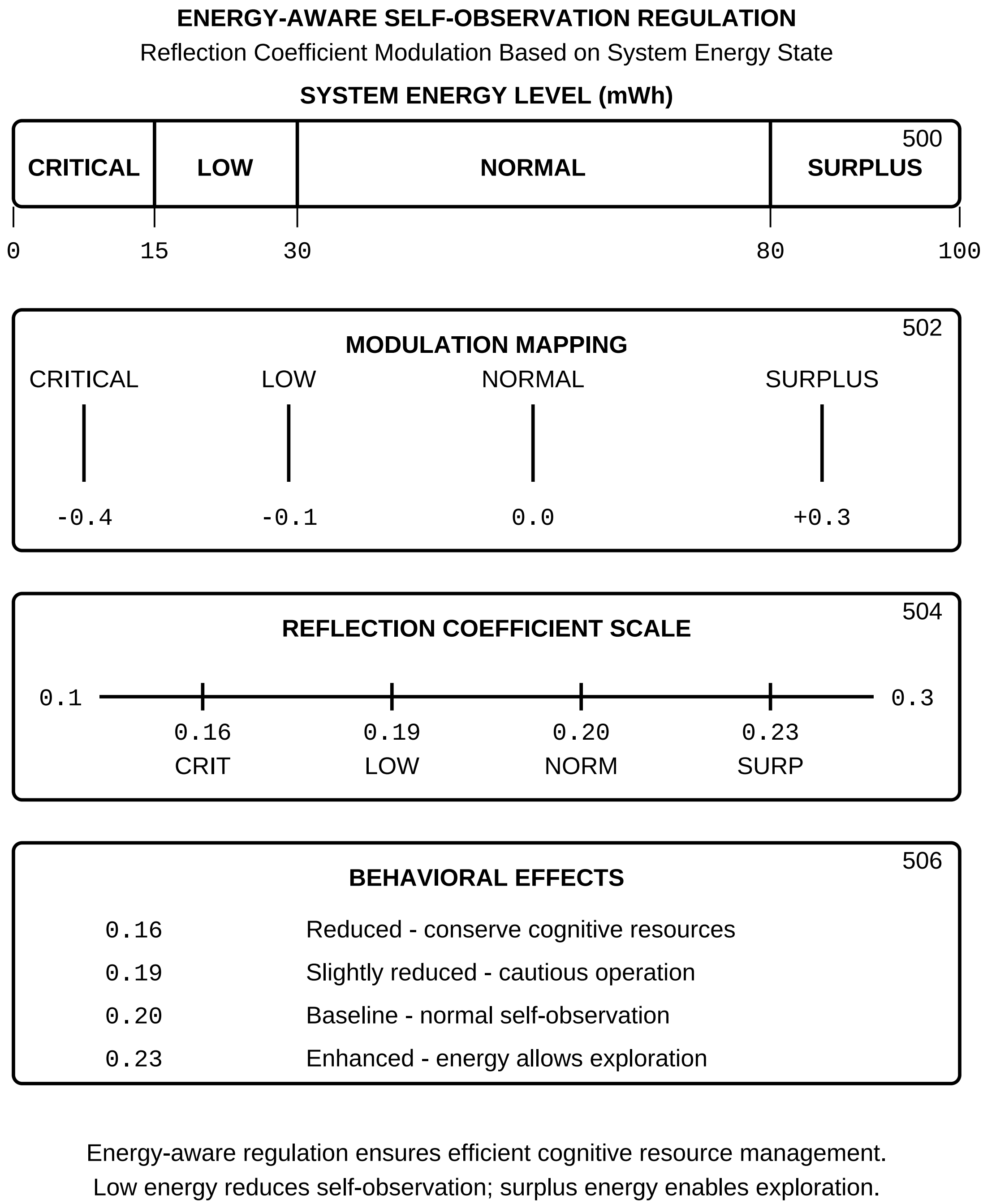
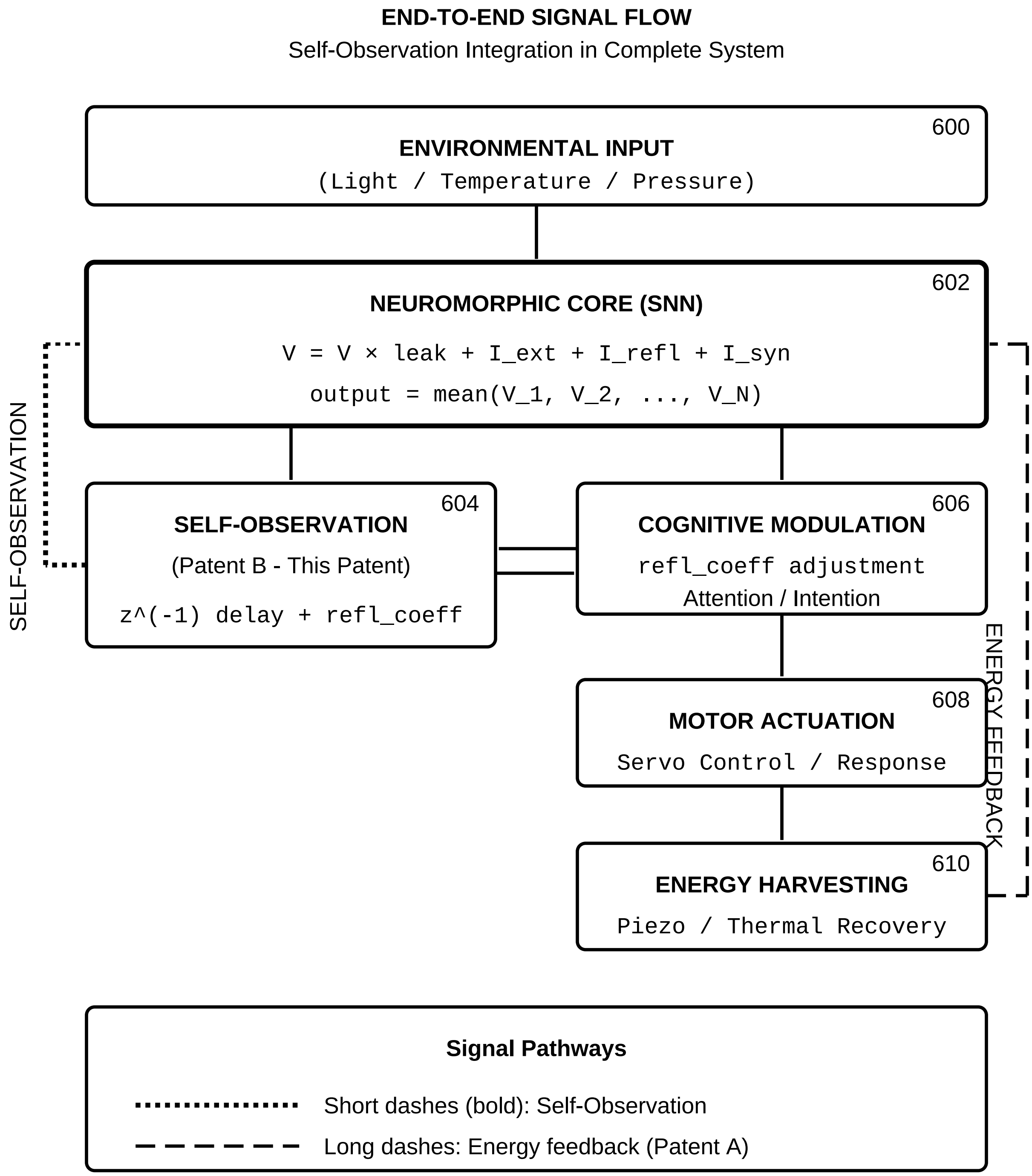

ENERGY-AWARE SELF-OBSERVATION REGULATION
Reflection Coefficient Modulation Based on System Energy State
SYSTEM ENERGY LEVEL (mWh)
500
CRITICAL
LOW
NORMAL
SURPLUS
0
15
30
80
100
502
MODULATION MAPPING
CRITICAL
LOW
NORMAL
SURPLUS
-0.4
-0.1
0.0
+0.3
504
REFLECTION COEFFICIENT SCALE
0.1
0.3
0.16
CRIT
0.19
LOW
0.20
NORM
0.23
SURP
506
BEHAVIORAL EFFECTS
0.16 Reduced - conserve cognitive resources
0.19 Slightly reduced - cautious operation
0.20 Baseline - normal self-observation
0.23 Enhanced - energy allows exploration
Energy-aware regulation ensures efficient cognitive resource management.
Low energy reduces self-observation; surplus energy enables exploration.

FIG. 5



The self-observation pathway feeds the network's own aggregate state back as explicit input at configurable gain (0.0 to 1.0).

FIG. 6